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Protein Language Model Identifies Disordered, Conserved Motifs Implicated in Phase Separation

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eLife Assessment

This **valuable** study presents an analysis of evolutionary conservation in intrinsically disordered regions, identified as key drivers of phase separation, leveraging a protein language model. The strength of evidence presented is **convincing** overall, though the theoretical grounding could benefit from further development.

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Abstract

Intrinsically disordered regions (IDRs) play a critical role in phase separation and are essential for the formation of membraneless organelles (MLOs). Mutations within IDRs can disrupt their multivalent interaction networks, altering phase behavior and contributing to various diseases. Therefore, examining the evolutionary constraints of IDRs provides valuable insights into the relationship between protein sequences and phase separation. In this study, we utilized the ESM2 protein language model to map the residue-level mutational tolerance landscapes landscape of IDRs. Our findings reveal that IDRs, particularly those actively participating in phase separation, contain conserved amino acids. This conservation is evident through mutational constraints predicted by ESM2 and supported by direct analyses of multiple sequence alignments. These conserved, disordered amino acids include residues traditionally identified as “stickers” as well as “spacers” and frequently form continuous sequence motifs. The strong conservation, combined with their potential role in phase separation, suggests that these motifs may act as functional units under evolutionary selection to support stable MLO formation. Our findings underscore the insights into phase separation’s molecular grammar made possible through evolutionary analysis enabled by protein language models.

Introduction

Membraneless organelles (MLOs), such as nucleoli, stress granules, and P-bodies, are distributed throughout diverse cellular environments and play a vital role in forming specialized biochemical compartments that drive essential cellular functions.[1–13] These biomolecular condensates typically assemble through phase separation, dynamically recruiting reactants and releasing products to improve the efficiency and specificity of cellular processes.[4,14,15] Intrinsically disordered regions (IDRs), which lack well-defined tertiary structures yet exhibit unique structural disorder, frequently act as scaffolds within MLOs.[14,16–19] IDRs facilitate multivalent interactions, including π-π stacking, cation-π, and electrostatic interactions, that promote phase separation. Mutations in IDRs can disrupt phase behavior, potentially leading to MLO dysfunction and contributing to diseases such as neurodegenerative disorders and cancer.[20–23]

Substantial research has focused on linking protein sequences to the phase behaviors of condensates.[4,6,7,9,17,24–42] These studies support the “stickers and spacers” framework,[3,4,43–46] in which specific residues, termed “stickers,” drive strong, specific interactions, while “spacer” regions act as flexible linkers with minimal nonspecific interactions.[37,47–50] Sood and Zhang [51] further introduced an evolutionary dimension, proposing that IDRs adapt this framework to balance effective phase separation with compositional specificity. For instance, membrane-less organelles (MLOs) form through specific interactions among stickers, while minimizing non-specific interactions among spacers to enable condensate formation with defined structural and compositional properties. This evolutionary pressure supports the enrichment of low-complexity domains (LCDs) in IDRs, reducing nonspecific interactions yet preserving conserved stickers critical for condensate specificity and stability. This evolutionary perspective, while promising, has not been extensively validated.

Evolutionary analysis of IDRs is challenging due to difficulties in sequence alignment,[52–58] though several studies have attempted alignment of disordered proteins with promising results.[59–61] Recent advances in protein language models[62–67] provide alternative approaches for sequence analysis and information decoding. Trained on large protein datasets, such as UniProt,[68] these models leverage neural network architectures like the Transformer[69] to capture correlations among amino acids within a sequence. These correlations enable the models to identify constraints, enforced by the whole sequence, that influence the chemical identity of amino acids at specific positions. Since these constraints often reflect functional or structural requirements, the mutational preferences predicted by these models can be interpreted as proxies for evolutionary constraint or mutational tolerance.[70–77] Unlike traditional methods, these predictions do not rely on sequence homology, making them particularly attractive for analyzing disordered protein sequences where alignment is unreliable.

While protein language models have been widely applied to structured proteins,[73–75,77–80] it is important to emphasize that these models themselves are not inherently biased toward folded domains. For example, the Evolutionary Scale Model (ESM2)[79] is trained as a probabilistic language model on raw protein sequences, without incorporating any structural or functional annotations. Its unsupervised learning paradigm enables ESM2 to capture statistical patterns of residue usage and evolutionary constraints without relying on explicit structural information. Thus, the success of ESM2 in modeling the mutational landscapes of folded proteins[73–75,77–79] reflects the model’s ability to learn sequence-level constraints imposed by natural selection — a property that is equally applicable to IDRs if those regions are also under functional selection. Indeed, protein language models are increasingly been used to analyze variant effects in IDRs.[70,71,78]

In this study, we employ the ESM2 model to investigate the mutational landscape of IDRs involved in MLO formation. Our analysis demonstrates the utility of ESM2 for examining disordered sequences, identifying a notable subset of amino acids that exhibit mutation resistance. These amino acids are evolutionarily conserved, as confirmed through multi-sequence alignment analysis. Notably, regions associated with phase separation are significantly enriched in these conserved residues. Importantly, the conserved disordered amino acids include both sticker residues, such as tyrosine (Y), tryptophan (W), and phenylalanine (F), and spacer residues, such as alanine (A), glycine (G), and Proline (P). These residues frequently colocalize within continuous sequence stretches, underscoring the functional relevance of entire motifs rather than isolated residues in MLO formation. Our findings provide strong evidence for evolutionary pressures acting on specific IDRs, likely to maintain their roles in scaffolding phase separation mechanisms.

Results

Protein Language Model for Quantifying the Mutational Landscape of MLO Proteins

To examine the mutational tolerance of IDRs and their connection to phase separation, we compiled a database of human proteins with disordered regions. From this, we identified a subset of 939 proteins associated with the formation of membraneless organelle, referred to as MLO-hProt. The Methods section provides additional information on the dataset preparation. The dataset contains proteins with varying numbers of disordered residues, ranging from a few dozen to several thousand per protein (**Figure 1A**). These proteins are involved in the assembly of various MLOs, including P-bodies, Cajal bodies, and centrosome granules, and are distributed across both nuclear and cytoplasmic compartments.

We analyzed proteins in the MLO-hProt dataset using the protein language model, ESM2.[79] As illustrated in **Figure 1B**, ESM2 is a conditional probabilistic model (masked language model) that predicts the likelihood of specific amino acids appearing at a given position, based on the surrounding sequence context.

The mutational tolerance of a specific amino acid at a given site is defined as follows. ESM2 enables the quantification of the probability, or likelihood, of observing any of the 20 amino acids at site i. To assess the preference for a mutant over the wild-type (WT) residue, we calculate the log-likelihood ratio (LLR) between the mutant and WT residues. Consequently, a 20-element vector representing the LLRs for each amino acid can be generated at each site (**Figure 1B**). This vector is then condensed into a single value, referred to as the ESM2 score, which is derived using an information entropy expression for the LLR probabilities of individual amino acids (**Eq. 1** in the Methods Section).

The ESM2 score provides a measure of the overall mutational tolerance of a given residue. Lower scores indicate higher mutational constraint and reduced flexibility, implying that these residues are more likely essential for protein function, as they exhibit fewer permissible mutational states.

ESM2 Identifies Conserved, Disorderd Residues

We next used ESM2 to analyze the mutational tolerance of amino acids in both structured and disordered regions. We carried out ESM2 predictions for all proteins in the MLO-hProt dataset, and determined the ESM2 scores of individual amino acids. In addition, to quantify structural disorder, we computed the AlphaFold2 predicted Local Distance Difference Test (pLDDT) scores for each residue. The pLDDT scores have been shown to correlate well with protein flexibility and disorder,[81] making them a reliable tool for distinguishing structured from unstructured regions.

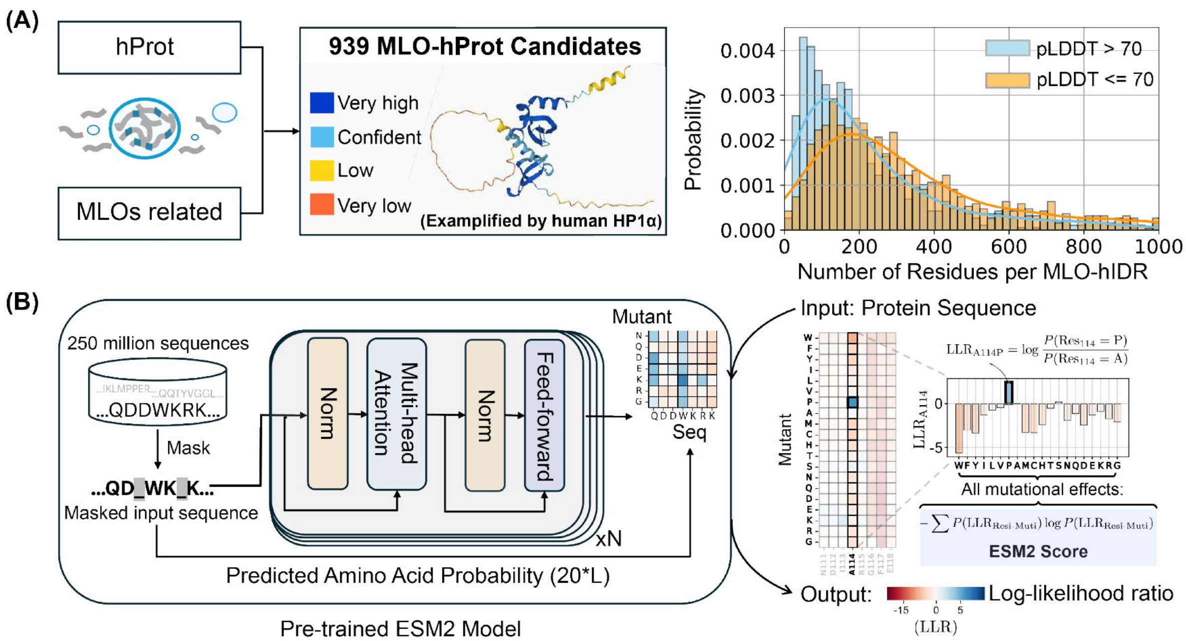

Figure 1

Predicting the mutational landscape of IDRs involved in MLO formation (MLO-hProt) using ESM2.

(A) Schematic representation of the MLO-hProt database construction. The right panel shows the distribution of disordered and folded residues identified in each protein, where structural order is assigned using the AlphaFold2 pLDDT score.(B) Workflow of the pre-trained ESM2 model (left)[79] for predicting the mutational landscape of a given protein sequence (right). Upon receiving a protein sequence as input, ESM2 generates a log-likelihood ratio (LLR) for each mutation type at each residue position. Using the 20-element LLR vector, we compute the ESM2 score for each residue (Eq. 1) to assess mutational tolerance.

Following previous studies,[82,83] we used a threshold of pLDDT = 70 to differentiate ordered from disordered residues. This threshold reflects amino acid composition preferences for folded versus disordered proteins (see Figure S1).[82,83]

We first analyzed the relationship between ESM2 and pLDDT scores for human Heterochromatin Protein 1α (HP1α, residues 1–191). HP1α is a crucial chromatin organizer that promotes phase separation and facilitates the compaction of chromatin into transcriptionally inactive regions.[34,84–87] HP1α comprises both structured and disordered segments, as illustrated in **Figure 2A**. Here, residues with pLDDT scores exceeding 70 (indicating ordered regions) are shown in white, while disordered segments (pLDDT ≤ 70) are highlighted in blue. Similar plots for other proteins are included in Figure S2. **Figure 2B** displays the AlphaFold2 predicted structure of HP1α, with residues colored according to their pLDDT scores.

Our analysis demonstrated that HP1α's structured domains consistently yield low ESM2 scores, reflecting strong mutational constraints characteristic of folded regions. These constraints are further evident in the local LLR predictions, as shown in **Figure 2B**, where we illustrate the folded region G120-T130. Given the critical role of maintaining the 3D conformation of structured domains, mutations with pronounced deleterious effects are likely to significantly disrupt protein folding. This interpretation aligns with prior studies that have reported a strong correlation between ESM2 LLRs and changes in free energy associated with protein structural stability.[78,79]

In contrast, disordered regions, including the N-terminal (residues 1–19), C-terminal (175–191), and hinge domain (70–117), typically exhibit higher ESM2 scores, indicating increased mutational flexibility. Nonetheless, not all disordered regions show similar flexibility. Within the hinge domain, a conserved segment known as the KRK patch (highlighted in orange)[84,88–90] shows low ESM2 scores, despite being disordered. This distinction allows us to classify disordered regions into two types: "flexible disordered" regions, which show high ESM2 scores and greater mutational tolerance, and "conserved disordered" regions, which display low ESM2 scores, indicating varying levels of mutational constraint despite a lack of stable folding.

We then examined the distribution of ESM2 scores for all amino acids in the MLO-hProt dataset to evaluate the generality of these patterns. Amino acids in folded regions (pLDDT > 70) consistently yield low ESM2 scores, reflecting strong mutational constraints. As shown in **Figure 2C**, the histogram of ESM2 versus pLDDT scores for structured residues reveals a dominant population with low ESM2 values (region **a**, ESM2 Score ≤ 0.5), consistent with the established understanding that folded domains require structural and functional integrity and are thus more mutation-sensitive.[91–93]

In contrast, disordered residues (pLDDT ≤ 70) predominantly show high ESM2 scores (region **II**, ESM2 Score ≥ 2.0), consistent with the rapid evolution and higher mutational tolerance typical of disordered proteins.[94–96] However, as shown in **Figure 2D**, a substantial subset of disordered amino acids also exhibit low ESM2 scores (region **I**). Given that low ESM2 scores generally reflect mutational constraint in folded proteins, the presence of region **I** among disordered residues suggests that certain disordered amino acids are evolutionarily conserved and likely functionally significant.

ESM2 Scores Correlate with Sequence Conservation

Our analysis indicates that a substantial proportion of amino acids within disordered regions exhibit low mutational tolerance. To evaluate this hypothesis, we conducted an evolutionary analysis of MLO-hProt proteins, examining the conservation patterns of individual amino acids.

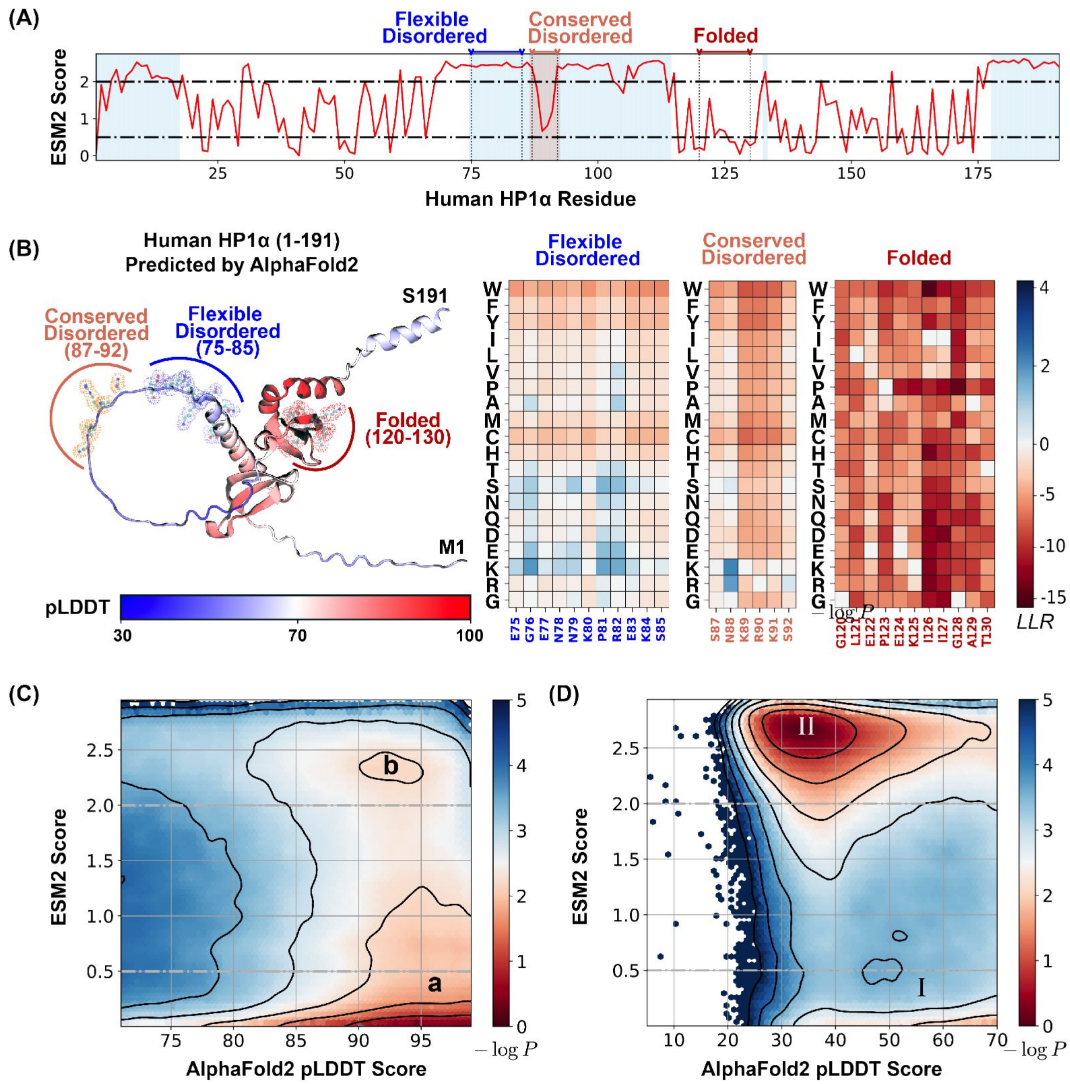

Figure 2

ESM2 predicts mutational landscape for structured and disordered residues.

(A) The ESM2 scores for amino acids in the human HP1α protein (UniProt ID: P45973) are presented, with residues having pLDDT scores below 70 highlighted in blue to signify regions lacking a defined structure. (B) A detailed view of the mutational landscape across three regions with varying degrees of structural order. On the left, the AlphaFold2-predicted structure of the human HP1α protein is displayed in cartoon representation, with residues colored according to their pLDDT scores. Three specific regions, representing flexible disordered (residues 75–85), conserved disordered (residues 87–92), and folded (residues 120–130) segments, are highlighted in blue, orange, and red, respectively, using ball-and-stick styles. The panels on the right depict the ESM2 LLR predictions for each of these regions. (C, D) Histograms of pLDDT and ESM2 score distributions for structured (C) and disordered (D) residues are presented. Contour lines indicate free energy levelscomputed as$-\log P$ (pLDDT, ESM2), where P is the probability density of residues based on their pLDDT and ESM2 scores. Contours are spaced at 0.5-unit intervals to distinguish areas of differing density.

This analysis was based on a multi-sequence alignment (MSA) of homologs of MLO-hProt proteins. We employed HHblits for homolog detection, a method particularly suited to disordered proteins as it effectively captures distant sequence similarities in highly divergent sequences.[97–99] The presence of folded domains in these proteins facilitates reliable alignment between references and their query homologs. To exclude sequences that no longer qualify as homologs, we filtered for sequences with at least 20% identity to the reference, resulting in homologous sets ranging from tens to thousands per protein (**Figure 3A**). From these aligned sequences, we calculated the conservation score for each reference amino acid as the ratio of its occurrence in homologs to the total number of sequences (see **Eq. 2** in the Methods Section).

Our findings reveal a strong correlation between ESM2 scores and conservation scores. In **Figure 3B**, we present the histograms of ESM2 and conservation scores for all amino acids from MLO-hProt proteins. Given that folded domains generally show higher conservation scores than disordered regions, we further classified residues into four groups based on their AlphaFold2 pLDDT scores to assess conservation patterns across varying levels of structural disorder. This stratification allowed us to analyze conservation trends in detail. Across all categories, we observed bimodal distributions, reinforcing the correlation between increasing ESM2 scores and decreasing conservation.

The conservation of amino acids with low ESM2 scores is also apparent in **Figure 3C**. For each of the four structural order groups, we computed the average ESM2 and conservation scores for all 20 amino acid types. Methionine (M) was excluded from the correlation analysis due to its frequent position as the initial residue in sequences, which complicates reliable mutational effect predictions.[70,100] In each group, we consistently observed a strong correlation between average ESM2 and conservation scores.

While ESM2 scores align closely with conservation scores, the relative conservation of specific amino acids varies across structural order groups. In more disordered regions, hydrophilic residues such as Glutamine (Q), Lysine (K), and Arginine (R) exhibit lower ESM2 scores, indicating that mutations in these residues are particularly detrimental. Conversely, hydrophobic residues like Valine (V) and Isoleucine (I) show higher ESM2 scores, suggesting they experience reduced evolutionary constraints. In more folded domains, hydrophobic residues such as W and F are more conserved (see Figure S3), consistent with the characteristic conservation patterns of proteins across different disorder levels.[101,102] Overall, these findings strongly support our hypothesis that ESM2 scores effectively capture evolutionary conservation, enabling the identification of functionally significant residues through the mutational landscape, independent of structural flexibility.

Regions Driving Phase Separation Are Enriched with Conserved, Disordered Residues

The presence of evolutionarily conserved disordered residues raises the question of their functional significance. To explore this, we identified disordered regions of MLO-hProt using a pLDDT score ≤ 70 and partitioned these regions into two categories: drivers (dMLO-hIDR), which actively drive phase separation, and clients (cMLO-hIDR), which are present in MLOs under certain conditions but do not promote phase separation themselves.[103] Additionally, IDRs from human proteins not associated with MLOs, termed nMLO-hIDR, were included as a control. To enhance statistical robustness, we extended our dataset by incorporating driver proteins from additional species,[104] resulting in the expanded dMLO-IDR dataset. Figure S4 shows the amino acid composition across these datasets. Beyond the pLDDT-based classification, the majority of residues in these datasets are also predicted to be disordered by various computational tools and supported by experimental evidence (Figures S5 and S6).

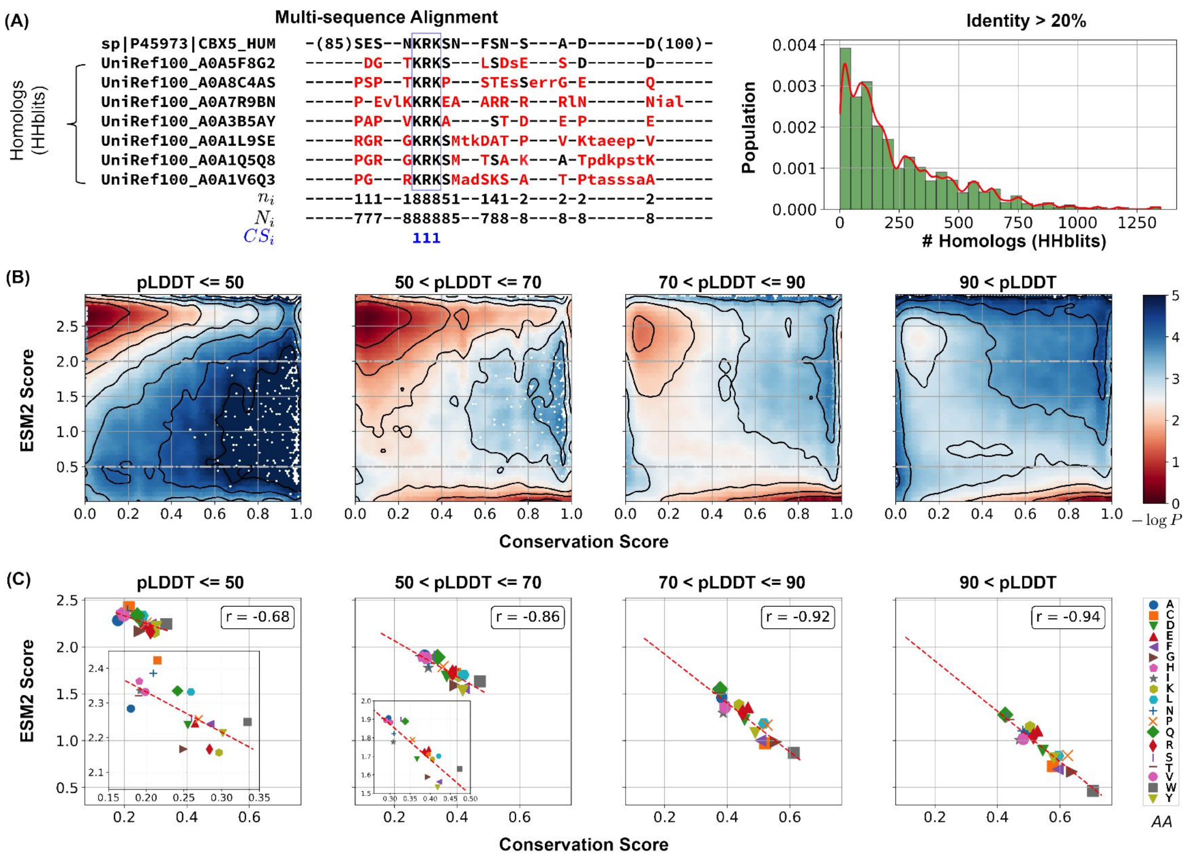

Figure 3

Low ESM2 scores correlate strongly with evolutionary conservation.

(A) Estimating amino acid conservation using multiple sequence alignment. The conservation score calculation is demonstrated for human HP1α protein along with a subset of its homologs found by HHblits. In the aligned sequences, missing residues appear as dashed lines, insertions are shown in lowercase letters, and mismatches are highlighted in red. The three rows below the alignment indicate at position i: n_i, the number of conserved residues from the reference sequence to the query sequences; N_i; the total number of existing residues; and CS_i, the conservation score calculated from **Eq. 2**, respectively. The right panel illustrates the distribution of homolog counts for each MLO-hProt found by HHblits. (B) Histograms showing conservation and ESM2 score distributions for all residues in MLO-hProt, grouped by pLDDT scores from AlphaFold2. The contour lines denote free energy levels, calculated as − log P (CS, ESM2), where P is the probability density of residues based on their con-servation and ESM2 scores. Contours are spaced at 0.5-unit intervals to highlight regions of distinct density. (C) Correlation between mean conservation and ESM2 scores for amino acids classified by structural order levels. Pearson correlation coefficients, r, are reported in the legends.

As illustrated in **Figure 4A**, there is a progressive increase in the fraction of conserved disordered residues and a corresponding decline in flexible disordered residues from non-phase-separating proteins (nMOL-hIDR) to clients (cMLO-hIDR) and drivers (dMLO-hIDR and dMLO-IDR) (see also Figure S7). Driver proteins, particularly those in the expanded dataset, display a notable reduction in flexible residues. These findings imply that disordered regions with a role in phase separation tend to contain functionally significant and evolutionarily conserved regions.

We further examined the sequence location of conserved, disordered residues in driver proteins (dMLO-IDR). For these proteins, experimentally verified segments have been identified, the deletion or mutation of which impairs phase separation[104–108] (**Figure 4B**). These segments can include both structured and disordered regions. Herein, if a disordered region constitutes over 50% of the phase-separating segment, we designate it as "driving", indicating a likely critical contribution to phase separation. If the disordered region represents less than 50%, we classify it as "participating", with a potentially limited role. Finally, if there is no overlap between the disordered region and the phase-separating segment, we categorize it as "non-participating". The number of segments in the three IDR groups, along with their amino acid compositions, are shown in Figure S8.

We then analyzed the distribution of ESM2 predictions across these IDR groups. In alignment with **Figure 3A**, we observed a significantly higher proportion of conserved disordered residues within driving IDRs, while few were present in non-participating IDRs. Supporting the ESM2 predictions, conservation analysis based on MSA also indicated that drving IDRs contain a greater concentration of conserved residues (**Figure 4C**). Collectively, these findings demonstrate that ESM2 effectively identifies evolutionarily conserved functional sites, enriched in IDR regions likely involved in driving phase separation.

Conserved, Disordered Residues Form Motifs

Finally, we investigated the chemical identities of conserved residues within driving IDRs to understand their potential role in phase separation. **Figure 5A** displays the average ESM2 LLR predictions for each of the 20 amino acids in the mutational matrix, indicating that mutations to most amino acids are generally unfavorable, as reflected by their low, negative LLR values. This trend is particularly pronounced in driving IDRs compared to nMLO-IDRs or non-participating IDRs (Figure S9).

We further characterize these conserved residues within driving IDRs. Using hierarchical clustering on two UMAP-derived embeddings from the LLR vectors, we grouped amino acids into five clusters (**Figure 5B**). This approach distinguishes more conserved residues (Groups I to III) from the more flexible residues (Groups IV and V). Notably, W, F, and Y—often referred to as "stickers" due to their crucial role in phase separation[43,109–111]—are uniquely grouped within the highly conserved Group I. These findings support the expectation that amino acids essential to phase separation are often evolutionarily conserved, aligning with their central role in functional stability.

Interestingly, residues in Groups II and III, which include traditional "spacers" (G, A, P, and S), also show high conservation and resistance to most mutation types, particularly hydrophobic mutations (**Figure 5A**). Spacer residues, generally regarded as less critical for interactions driving condensate formation, were unexpectedly conserved, suggesting a broader functional relevance than previously assumed.

We propose that this conservation pattern for spacers is likely not due to isolated residue conservation but may instead reflect the conservation of specific sequence stretches. To examine this hypothesis, we identified ESM2 "motifs" as continuous sequence regions with average ESM2 scores below 0.5. A full list of motifs is available in the appendix (ESM2_motif_with_exp_ref.csv). We observed that conserved amino acids with ESM2 scores below 0.5 are predominantly located

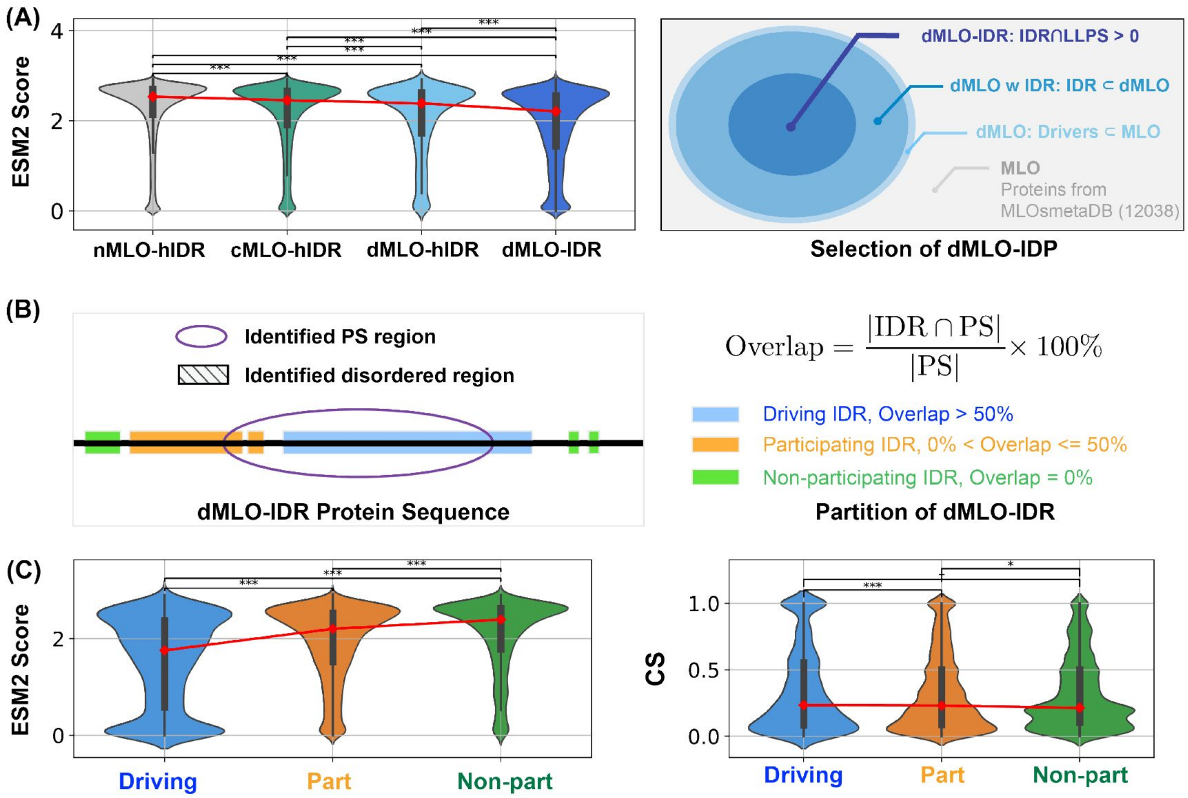

Figure 4

Phase separation driving IDRs exhibit more conserved disorder.

(A) Population of ESM2 score for disordered residues in proteins from nMLO-hIDR, cMLO-hIDR, dMLO-hIDR, and dMLO-IDR datasets. Red dots indicate the mean values of the respective distributions. The selection of proteins in the dMLO-IDR dataset is shown in the right panel. See also methods for details in dataset preparation. (B) The classification of three IDR functional groups based on their overlap with the experimentally identified phase separation (PS) segments. (C) The distribution of the ESM2 score for residues in three IDR groups, driving (blue), participating (orange), and non-participating (green) shown in the violin plot. (D) The distribution of the conservation score (CS) for residues in three IDR groups shown in the violin plot with same coloring scheme in C. Pairwise statistical comparisons were conducted using two-sided Mann–Whitney U tests on the ESM2 score distributions (null hypothesis: the two groups have equal medians). P-values indicate the probability of observing the observed rank differences under the null hypothesis. Statistical significance is denoted as follows: *******: $p < 0.001$; ******: $p < 0.01$; *****: $p < 0.05$; **†**: $p < 0.10$ (marginal); **n.s.**: (not significant, $p \geq 0.10$).

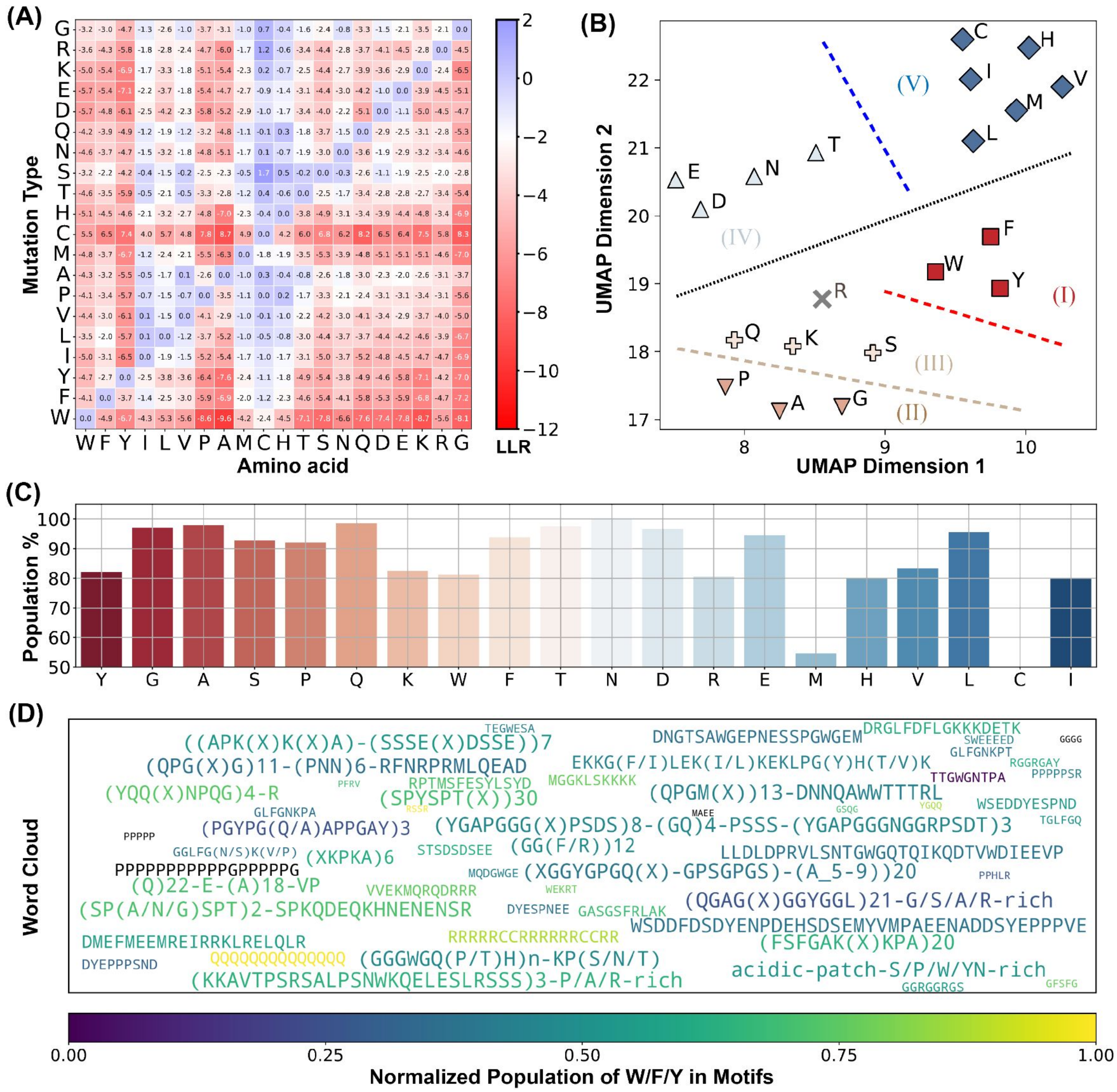

Figure 5

ESM2 Identifies conserved Motifs in driving IDRs.

(A) Mean LLR values for the 20 amino acids calculated by averaging across all residues of each amino acid type. (B) Clustering of amino acids based on the two UMAP embeddings of their LLR vectors presented in part A. The dashed lines are manually added for a clear visualization of the separation of each group. (C) The percent of conserved residues locating in motifs for all amino acids. (D) Word cloud of motifs identified by ESM2. The word font size reflects the relative motif length, while the color represents the proportion of “sticker” residues (Y, F, W, R, K, and Q) within each motif.

within these motifs (see **Figure 5C** and Figure S10A). For instance, conserved glycine residues have a 97.9% likelihood of being part of an ESM2 motif, with similar probabilities observed for other spacers, such as alanine (99.0%) and proline (93.1%), as well as for sticker residues like Y, W, and F.

These results suggest that IDRs crucial for phase separation frequently contain conserved sequence motifs composed of both sticker and spacer residues. Interestingly, many of thesemotifs have been experimentally validated as essential for phase separation, with representative motifs for each driving IDR listed in ESM2_motif_with_exp_ref.csv. In these cases, mutations or deletions have been shown to disrupt phase separation. For visualization, a word cloud of these motifs is presented in **Figure 5D**. Altogether, our analysis suggest a tendency for IDRs to uniquely cluster conserved residues into motifs and execute significant biological roles in phase separation.

Conclusions and Discussion

We have utilized the protein language model ESM2 to investigate the mutational landscape of IDRs. Our analysis reveals a substantial population of mutation-resistant amino acids. Multi-sequence alignment confirms their evolutionary conservation. Notably, regions actively involved in phase separation are enriched with these conserved, disordered residues, supporting their potential role in the formation of membraneless organelles. These findings underscore evolutionary constraints on specific IDRs to preserve their functional roles in scaffolding phase separation processes.

We emphasize that the results presented in **Figure 4** do not directly demonstrate that conserved residues are preferentially located in regions associated with phase separation. Although these regions are more enriched in conserved amino acids, their total sequence length can be smaller than that of non-phase-separating regions. As a result, the absolute number of conserved residues may still be higher outside phase-separating regions. To quantitatively assess this, we calculated, for each protein in the MLO-hProt dataset, the probability p of finding conserved residues within regions contributing to phase separation. These regions include both “driving” and “participating” segments, as defined in **Figure 4** of the main text. Figure S11 shows the distribution of p across all proteins. For comparison, we also present the distribution of $1 - p$, which reflects the probability of finding conserved residues in non-phase-separating regions. While the average value of p exceeds 0.5, indicating a trend toward conserved residues being more frequently located in phase-separating regions, the difference between the two distributions is not statistically significant. Future studies with expanded datasets may be necessary to clarify this trend.

Related Work

Numerous studies have sought to identify functionally relevant amino acid groups within IDRs.[11,34,43,46,109,112–116] For instance, using multiple sequence alignment, several groups have identified evolutionarily conserved residues that contribute to phase separation.[13,59,61,84,113,117–121] Alderson et al. employed AlphaFold2 to detect disordered regions with a propensity to adopt structured conformations, suggesting potential functional relevance.[83]

In contrast, our approach based on ESM2 is more direct: it identifies conserved residues without relying on alignment or presupposing that functional significance requires folding into stable 3D structures. Notably, many of the conserved residues identified in our analysis exhibit low pLDDT scores (**Figure 2**), implying potential functional roles independent of stable conformations.

Relationship to the Stickers and Spacers Model

To interpret the functional implications of conserved residues, we contextualize our findings within the stickers-and-spacers framework. Our results indicate that these conserved amino acids encompass both "sticker" and "spacer" classifications, as defined in recent literature.[13,84,111,113,117–119,121–123] This suggests that residue-level classification may obscure functional roles in MLO formation.

Instead, evolutionarily conserved motifs, which can be readily identified through ESM2 score profiles, likely represent functionally significant units that integrate combinations of stickers and spacers. Perturbing these motifs in in vivo systems may offer a productive strategy for elucidating their biological functions.

Outside the conserved motifs, IDRs display greater mutational tolerance, with a general preference for hydrophilic residues. This structural arrangement—functionally constrained motifs embedded in more flexible, mutation-tolerant regions—is consistent with a generalized stickers-and-spacers framework. Conserved motifs likely engage in stronger interactions that drive phase separation, while interactions among non-motif regions remain attenuated. This spatial organization supports the formation of MLOs with defined chemical compositions and interaction specificity, as proposed by Sood and Zhang [51].

Future Directions

Several promising directions could extend this work, both to refine our mechanistic understanding and to explore clinical relevance. One avenue is testing the hypothesis that conserved motifs in scaffold proteins act as functional stickers, mediating strong intermolecular interactions. This could be evaluated computationally via free energy calculations or experimentally via interaction assays. Deletion of such motifs in client proteins may also reduce their partitioning into condensates, illuminating their roles in molecular recruitment.

To explore potential clinical implications, we analyzed pathogenicity data from ClinVar.[124,125] As shown in Figure S12A, single-point mutations with low LLR values—indicative of constrained residues—are enriched among clinically reported pathogenic variants, while benign variants typically exhibit higher LLR values. Moreover, mutations within conserved motifs are significantly more likely to be pathogenic than those in non-motif regions (Figure S12B). These findings highlight the potential of ESM2 as a first-pass screening tool for identifying clinically relevant residues and suggest that the conserved motifs described here may serve as priorities for future studies, both mechanistic and therapeutic.

Limitations

Despite these promising findings, our study has several limitations. Most notably, the analysis is entirely computational, relying on ESM2-derived predictions and sequence-based conservation without direct experimental validation. While the strong correlation between ESM2 scores and evolutionary conservation suggests functional constraint, the specific biological roles of many conserved motifs remain uncharacterized.

The motifs we identify are located within broader regions that have previously been shown, through experimental studies, to drive phase separation. This provides circumstantial support for their functional relevance. However, these motifs typically represent only a small fraction of the larger disordered segments implicated in phase separation. As such, it is not possible to attribute the functional effect of the entire region solely to the identified motifs—other segments of the sequence may also make essential contributions. Therefore, while suggestive, the current evidence does not constitute direct experimental validation of motif-level functionality.

Future studies should aim to dissect the roles of these motifs in isolation and in context, using targeted mutagenesis, biophysical assays, and cellular perturbations. Such work will be crucial for establishing whether the conserved motifs identified here act as discrete functional units or as components of a broader sequence context required for biomolecular condensation and related processes.

Methods

Data Collection and Preprocessing

Human protein dataset

To construct the MLO-hProt dataset, we initially identified human proteins with disordered residues using the UniProtKB database.[126] From these proteins, we selected candidates with at least 10% of their residues exhibiting a pLDDT score ≤ 70, yielding a total of 5,121 candidates. These were then cross-referenced with entries in the CD-code Database,[127] yielding a final subset of 939 proteins associated with membraneless organelle formation (**Figure 1A**). The remaining 4,182 proteins, which were not linked to membraneless organelle formation, comprise the nMLO-hProt dataset. We further identified disordered regions from the datasets by selecting residues with pLDDT score ≤ 70, refereed as MLO-hIDR and nMLO-hIDR. The CD-code Database also classifies the MLO-hProt proteins into two categories: drivers (n = 82) and clients (n = 814), with the corresponding disordered regions labeled as dMLO-hIDR and cMLO-hIDR. Additional information on MLO-hIDR candidate proteins and their biological roles is available in the appendix file MLO-hIDR.csv.

dMLO-IDR dataset

In addition to datasets comprising only human proteins, we developed a specialized dMLO-IDR database incorporating proteins from diverse species involved in driving phase separation. This database includes all Driver proteins cataloged in the MLOsMetaDB database,[104] which documents experimentally validated disordered and phase-separating regions. Beginning with 780 Driver proteins, we filtered the dataset to retain entries where disordered regions overlap with phase-separating segments, yielding 399 candidates across 40 species (**Figure 4A**). To further refine the dataset, we analyzed sequence identity, excluding homologous pairs with sequence identity exceeding 50%. This process resulted in a final set of 341 non-redundant candidates. These proteins play critical roles in mediating the formation of various MLOs, including P-bodies, stress granules, paraspeckles, and centrosomes (see Appendix: dMLO-IDR.csv).

AlphaFold2 score for structural order

The pLDDT scores for human proteins were retrieved from the AlphaFold Protein Structure Database[128] by selecting the *Homo sapiens* organisms (Reference Proteome ID: UP000005640).

ESM2 predictions for Mutational Preferences

We employed the code and pretrained parameters for ESM2 available from the model's official GitHub repository at *https://github.com/facebookresearch/esm* to conduct mutational predictions. To optimize computational efficiency, we utilized the esm2_t33_650M_UR50D model, which has

650 million parameters and achieves prediction accuracy comparable to the larger 15B parameter model.[78] In addition to calculating the log-likelihood ratio (LLR) for individual mutations, we defined an ESM2 score at each position to quantify mutational tolerance, formulated as

$$\text{ESM2 Score} = -\sum_{i=1}^{20}\left(\frac{e^{\text{LLR}_i}}{\sum_{j=1}^{20} e^{\text{LLR}_j}} \log\left(\frac{e^{\text{LLR}_i}}{\sum_{j=1}^{20} e^{\text{LLR}_j}}\right)\right) \tag{1}$$

where LLR_i denotes the LLR value for the i-th amino acid mutation type.

Evolutionary Sequence Analysis

We performed multi-sequence alignment (MSA) analysis using HHblits from the HH-suite3 software suite,[129–132] a widely used open-source toolkit known for its sensitivity in detecting sequence similarities and identifying protein folds. HHblits builds MSAs through iterative database searches, sequentially incorporating matched sequences into the query MSA with each iteration. Sequence alignment was performed using the full-length protein sequences, encompassing both folded and disordered regions.

The HH-suite3 software was obtained from its GitHub repository (*https://github.com/soedinglab/hh-suite*). Homologous sequences were identified through searches against the UniRef30 protein database (release 2023/02).[133] For each query, we performed three iterations of HHblits searches, incorporating sequences from profile HMM matches with an E-value threshold of 0.001 into the query MSA in each cycle. Using a lower E-value threshold (closer to 0) ensures greater sequence similarity among the matches, while multiple iterations enhance the alignment's depth and accuracy. The resulting alignments in A3M format were converted to CLUSTAL format using the reformat.pl script provided in HH-suite, aligning all sequences to a uniform length (**Figure 3A**).

To refine alignment quality by focusing on closely related homologs, we filtered out se-quences with ≤ 20% identity to the query, excluding weakly related sequences where only short segments show similarity to the reference. For each sequence, we calculated the percent identity by counting the number of positions, denoted as n, at which the amino acid matches the reference. The percent identity was then computed as n/N, where N represents the total length of the aligned reference sequence. This total includes residues in folded and disordered regions, as well as gap positions introduced during alignment.

The conservation score CS_i for position i was then calculated from the MSA as

$$\text{CS}_i = \frac{n_i(\text{ref=query})}{N_i(\text{non\_gap})}, \tag{2}$$

where n_i(ref=query) represents the number of times the residues from the reference sequence appear across all sequences, and N_i(non_gap) represents the total non-gap residues across the aligned sequences.

Motif Identification

We defined motifs as contiguous stretches of amino acid sequences with an average ESM2 score of 0.5 or lower. To identify motifs within a given IDR, we implemented the following iterative procedure. Starting from either the N– or C–terminus of the sequence, we first locate the initial residue i whose ESM2 score falls within 0.5. From i, residues are sequentially appended in the direction toward the opposite terminus until the segment's average ESM2 score exceeds 0.5; the first residue to breach this threshold is denoted j. The segment (i, $i+1$, …, $j-1$) is then recorded as a candidate motif. This process repeats starting from j until the end of the IDR is reached.

We perform this full procedure independently from both termini and designate the final motif as the intersection of the two candidate-motif sets. This bidirectional overlap strategy excludes terminal residues that might transiently satisfy the average-score criterion only due to adjacent low-scoring regions, thereby isolating the conserved core of each motif. All other residues—those not included in either directional pass—are classified as non-motif regions, minimizing peripheral artifacts.

When two motifs are in close proximity along the sequence, they may be merged into a single motif. Specifically, if the starting position of one motif is within 8 residues of the ending position of another, we define a candidate segment as the sequence spanning both motifs and the intervening residues. If the candidate segment's average ESM2 score is below 0.5, it is included as a merged motif, replacing the individual motifs in the final list (Appendix: ESM2_motif_with_exp_ref.csv). In the analyses shown in **Figure 5**, we showed all motifs with $n \geq 4$; however, varying motif minimal length n does not alter the overall conclusions (Figure S10B).

Data and materials availability

https://github.com/ZhangGroup-MITChemistry/ESM2_IDR_LLPS

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Additional files

Supplementary Information

Additional information

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Reviewer #1 (Public review):

The manuscript by Zhang et al describes the use of a protein language model (pLM) to analyse disordered regions in proteins, with a focus on those that may be important in biological phase separation. This is an interesting study that supports, complements and extends previous related analyses on the conservation and mutational tolerance of disordered regions, with a particular focus on disordered regions in proteins that are found in condensates.

https://doi.org/10.7554/eLife.105309.2.sa2

Reviewer #2 (Public review):

This manuscript uses the ESM2 language model to map the evolutionary fitness landscape of intrinsically disordered regions (IDRs). The central idea is that mutational preferences predicted by these models could be useful in understanding eventual IDR-related behavior, such as disruption of otherwise stable phases. While ESM2-type models have been applied to analyze such mutational effects in folded proteins, they have not been used or verified for studying IDRs. Here, the authors use ESM2 to study membraneless organelle formation and the related fitness landscape of IDRs.

Through this, their key finding in this work is the identification of a subset of amino acids that exhibit mutation resistance. Their findings reveal a strong correlation between ESM2 scores and conservation scores, which if true, could be useful for understanding IDRs in general. Through their ESM2-based calculations, the authors conclude that IDRs crucial for phase separation frequently contain conserved sequence motifs composed of both so-called sticker and spacer residues. The authors note that many such motifs have been experimentally validated as essential for phase separation.

Comments on revisions:

Unfortunately my concerns about lack of theoretical grounding and validation (especially critical in lack of theoretical grounding) persist. The argument about correlation between ESM2 scores and MSA conservation is circular. Protein language models already encode residue-level conservation, so agreement with conservation does not establish new predictive power. For IDRs, conservation is a poor surrogate for function because many functions are mediated by short, degenerate SLiMs that are frequently gained and lost. Sequence-only predictions therefore need orthogonal (preferably experimental or at the least in silico) tests. Finally, without a family-level holdout (e.g., cluster de-duplication at low identity) and prospective tests, overlap with known motifs cannot rule out training-data memorization/near-duplicates.

https://doi.org/10.7554/eLife.105309.2.sa1

Author response:

The following is the authors' response to the original reviews

> ***Reviewer #1:***
>
> *Summary:*
>
> *The manuscript by Zhang et al describes the use of a protein language model (pLM) to analyse disordered regions in proteins, with a focus on those that may be important in biological phase separation. While the paper is relatively easy to read overall, my main comment is that the authors could perhaps make it clearer which observations are new, and which support previous work using related approaches. Further, while the link to phase separation is interesting, it is not completely clear which data supports the statements made, and this could also be made clearer.*

We thank the reviewer for their thoughtful evaluation of our manuscript and for the supportive comments. As outlined in the responses below, we have made substantial revisions to clarify the novel observations presented in our study and to strengthen the connection between sequence conservation and phase separation.

> *Comment 1: With respect to putting the work in a better context of what has previously been done before, this is not to say that there is not new information in it, but what the authors do is somewhat closely related to work by others. I think it would be useful to make those links more directly.*

We have addressed the specific comments as outlined below.

> *Comment 1a: Alderson et al (reference 71) analysed in detail the conservation of IDRs (via pLDDT, which is itself related to conservation) to show, for example, that conserved residues fold upon binding. This analysis is very similar to the analysis used in the current study (using ESM2 as a different measure of conservation). Thus, the result that "Given that low ESM2 scores generally reflect mutational constraint in folded proteins, the presence of region a among disordered residues suggests that certain disordered amino acids are evolutionarily conserved and likely functionally significant" is in some ways very similar to the results of that (Alderson et al) paper .*

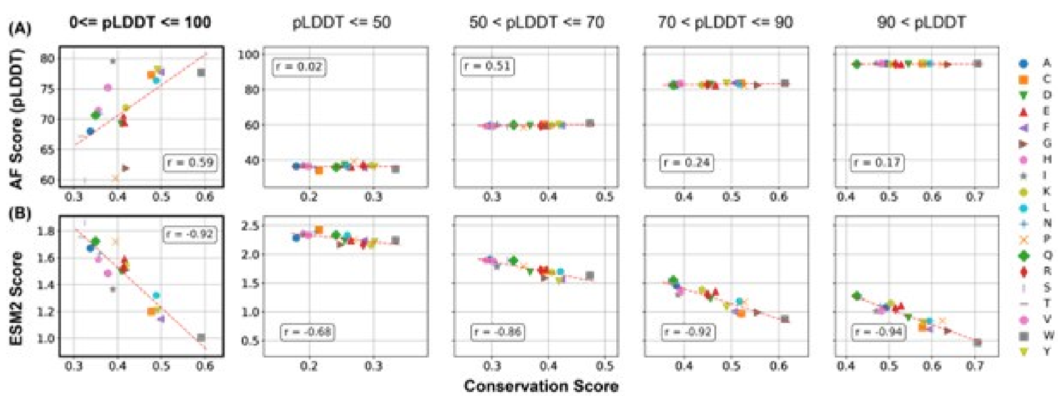
We thank the reviewer for the comment. However, we would like to clarify that our findings show subtle but important differences from those reported by Alderson et al. Specifically, Alderson et al. used AlphaFold2 predictions to identify IDRs that undergo disorder-to-order transitions, which the authors termed as conditionally folded IDRs. These regions could potentially be functionally important, assuming that function of IDRs necessitate folding.

We argue, however, that, the validity of this structure-function relationship for IDRs remains to be tested. In our opinion, The most direct way to evaluate the functional significance is via evaluating the evolutionary conservation.

As shown in Author response image 1, the correlation between pLDDT scores and the conservation score, while noticable, is significantly weaker than that between the ESM2 score and the conservation score.

Author response image 1.

Comparison of the correlation between AlphaFold2 pLDDT scores and conservation scores with the correlation between ESM2 scores and conservation scores. Calculations were performed using proteins in the MLO-hProt dataset. (A) Correlation between the mean AlphaFold2 pLDDT scores and conservation scores for various amino acids. Pearson correlation coefficients (*r*) are indicated in the figure legends. The four panels on the right present analogous correlation plots for amino acids grouped by structural order, as defined by their pLDDT scores. (B) Similar as in part A but for ESM2 scores.



Therefore, we believe that ESM2 score is a better indicator than AlphaFold2 pLDDT score for functional relevance.

Furthermore, for the human IDRs, we explicitly selected amino acids with pLDDT scores ≤ 70.

These would be classified as structureless, disordered amino acids, according to the study by Alderson et al. Nevertheless, as shown in Figures 2 and 3 of the main text, our analyses still identifies conserved regions. Therefore, these regions may function via distinct mechanisms than the disorder to order transition.

We now discuss the novelty of our work in the context of existing studies in the newly added Conclusions and Discussion: Related Work, as quoted below.

“Numerous studies have sought to identify functionally relevant amino acid groups within IDRs [cite]. For instance, using multiple sequence alignment, several groups have identified evolutionarily conserved residues that contribute to phase separation [cite]. Alderson et al. employed AlphaFold2 to detect disordered regions with a propensity to adopt structured conformations, suggesting potential functional relevance [alderson et al].

In contrast, our approach based on ESM2 is more direct: it identifies conserved residues without relying on alignment or presupposing that functional significance requires folding into stable 3D structures. Notably, many of the conserved residues identified in our analysis exhibit low pLDDT scores (Figure 2), implying potential functional roles independent of stable conformations.”

> *Comment 1b: Dasmeh et al, Lu et al and Ho & Huang analysed conservation in IDRs, including aromatic residues and their role in phase separation.*

We thank the reviewer for bringing these works to our attention! We now explicitly discuss these studies in both the Discussion section as mentioned above and in the Introduction as quoted below.

“Evolutionary analysis of IDRs is challenging due to difficulties in sequence alignment [cite], though several studies have attempted alignment of disordered proteins with promising results [Dasmeh et al, Lu et al and Ho & Huang].”

> *Comment 1c: A number of groups have performed proteomewide saturation scans using pLMs, including variants of the ESM family, including Meier (reference 89, but cited about something else) and Cagiada et al (https://doi.org/10.1101/2024.05.21.595203) that analysed variant effects in IDRs using a pLM. Thus, I think statements such as "their applicability to studying the fitness and evolutionary pressures on IDRs has yet to be established" should possibly be qualified.*

We added a new paragraph in the *Introduction* to discuss the application of protein language models to IDRs and cited the suggested references.

“While protein language models have been widely applied to structured proteins [cite], it is important to emphasize that these models themselves are not inherently biased toward folded domains. For example, the Evolutionary Scale Model (ESM2) [cite] is trained as a probabilistic language model on raw protein sequences, without incorporating any structural or functional annotations. Its unsupervised learning paradigm enables ESM2 to capture statistical patterns of residue usage and evolutionary constraints without relying on explicit structural information. Thus, the success of ESM2 in modeling the mutational landscapes of folded proteins [cite] reflects the model’s ability to learn sequence-level constraints imposed by natural selection — a property that is equally applicable to IDRs if those regions are also under functional selection. Indeed, protein language models are increasingly been used to analyze variant effects in IDRs [cite].”

> *Comment 2: On page 4, the authors write, "The conserved residues are primarily located in regions associated with phase separation." These results are presented as a central part of the work, but it is not completely clear what the evidence is.*

We thank the reviewer this insightful comment. We realized that our wording is not as precise as we should have been. We meant to state that the regions associated with phase separation are significantly enriched in these conserved residues. This is a significant finding and indicates that phase separation could be a source of evolutionary pressure in dictating IDP sequence conservation. However, we do not intend to suggest that phase separation is the only evolutionary pressure.

The sentence has been revised to

“Notably, regions associated with phase separation are significantly enriched in these conserved residues.”

We further replaced the section title "Conserved, Disordered Residues Localize in Regions Driving Phase Separation" with "Regions Driving Phase Separation Are Enriched with Conserved, Disordered Residues" to further clarify our findings and avoid overinterpretation.

Finally, we revised the following sentence in the discussion

“Notably, these conserved, disordered residues are predominantly located in regions actively involved in phase separation, contributing to the formation of membraneless organelles.”

to

“Notably, regions actively involved in phase separation are enriched with these conserved, disordered residues, supporting their potential role in the formation of membraneless organelles.”

The submitted manuscript provides clear evidence supporting the enrichment of conserved residues in MLO-driving IDRs. Specifically, Figures 4A and 4C demonstrate that these IDRs exhibit a substantially higher fraction of conserved residues compared to other IDRs involved in phase separation.

In this analysis, the nMLO-hIDR group serves as a baseline, representing the distribution of conservation in disordered regions lacking MLO-related functions. In contrast, IDRs from MLOassociated groups show a pronounced lower shift in their median and interquartile ranges, indicating stronger evolutionary constraints. Within the dMLO cohort, the degree of conservation follows a distinct gradient: driving residues exhibit the highest levels of conservation, followed by participant residues, with non-participant residues showing values closer to the nMLO baseline. This pattern reflects the relative functional importance of each group in phase separation, with conservation levels corresponding to their roles in MLO scaffolding.

To further support this, we computed, for each IDR, the fraction of conserved amino acids. As shown in Figure S11B, for IDRs that actively contribute to phase separation, the fraction is indeed higher than those not involved in phase separation. This analysis is now included in SI.

During the revision, we explicitly evaluated whether conserved residues are preferentially located in regions associated with phase separation. To this end, for each protein in the MLO-hProt dataset, we calculated the probability p of finding conserved residues within regions contributing to phase separation. These regions include both "driving" and "participating" segments as defined in Figure 4 of the main text.

Figure S11A presents the distribution of p across all proteins. For comparison, we also include the distribution of $1-p$, representing the probability of finding conserved residues in regions not associated with phase separation. On average, p exceeds 0.5, suggesting a tendency for conserved residues to be more frequently located in phase-separating regions. However, the difference between the two distributions is not statistically significant. This result may be due to the generally low density of conserved residues in IDRs, which makes the estimation of p challenging for individual proteins. Additionally, some conserved sites may be involved in functions unrelated to phase separation.

We added the following text to the *Discussion* section of the main text.

“We emphasize that the results presented in Figure 4 do not directly demonstrate that conserved residues are preferentially located in regions associated with phase separation. Although these regions are more enriched in conserved amino acids, their total sequence length can be smaller than that of non-phase-separating regions. As a result, the absolute number of conserved residues may still be higher outside phase-separating regions. To quantitatively assess this, we calculated, for each protein in the MLO-hProt dataset, the probability p of finding conserved residues within regions contributing to phase separation. These regions include both "driving" and "participating" segments, as defined in Figure 4 of the main text. Figure S11 shows the distribution of p across all proteins. For comparison, we also present the distribution of $1-p$, which reflects the probability of finding conserved residues in non-phase-separating regions. While the average value of p exceeds 0.5, indicating a trend toward conserved residues being more frequently located in phase-separating regions, the difference between the two distributions is not statistically significant. Future studies with expanded datasets may be necessary to clarify this trend."

> *Comment 3: It would be useful with an assessment of what controls the authors used to assess whether there are folded domains within their set of IDRs.*

We acknowledge that our previous labeling may have caused some confusion. Protein sequences used in Figures 2 and 3 include both folded and disordered domains. Results presented in these figures were constructed using full-length protein sequences to highlight the similarities and differences in ESM2 scores between folded and disordered domains.

In contrast, the analyses presented in Figures 4 and 5 focus exclusively on IDRs to examine their role in phase separation.

To prevent further confusion, we have renamed the dataset used in Figures 2 and 3 as MLO-hProt, emphasizing that the analysis pertains to entire protein sequences. The term MLO-hIDR is now reserved for a new dataset that includes only disordered residues, as used in Figures 4 and 5, and corresponding SI Figures.

For the dMLO-IDR dataset, all except one amino acid (P40967, residue G592) are annotated as disordered in the MobiDB database (https://mobidb.org/). This database characterizes disordered regions based on a combination of predictive algorithms and experimental data. As illustrated in Figure S5A, 25.5% of the proteins in the dataset have direct experimental evidence supporting their disorderedness. These experimental annotations are derived from a diverse range of techniques (Figure S5B). For the remaining proteins, disorder was predicted by one or more computational tools. Although not all tools were applied to every protein, each protein in the dataset was identified as disordered by at least one method.

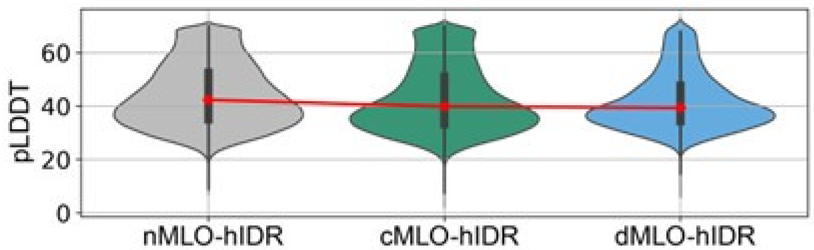
For human proteins, IDRs were identified based on AlphaFold2 pLDDT scores, using a threshold of 70. As established in prior studies [1, 2], the pLDDT score provides a quantitative measure of local structural confidence, with lower values indicating greater structural disorder. IDRs associated with conditional folding or disorder-to-order transitions generally exhibit high pLDDT values (e.g., >70).

Author response image 2 shows a violin plot of AlphaFold2 pLDDT scores for the various MLO-hIDR groups. The consistently low scores support the conclusion that these regions are structurally disordered.

We also cross-checked the MLO-hIDR regions against the MobiDB database. As shown in Figure S6, approximately 76% of the proteins in the dataset are predicted to contain disordered regions. Among the non-labeled segments with pLDDT scores ≤ 70, the majority are relatively short, with segments of 1–5 residues accounting for approximately 80%.

Author response image 2.

AlphaFold pLDDT scores of hIDRs in different MLO-related groups.



In addition to renaming the dataset, we also revised the manuscript to highlight the validation of disorderedness in section of *Results: Regions Driving Phase Separation Are Enriched with Conserved, Disordered Residues*.

“The presence of evolutionarily conserved disordered residues raises the question of their functional significance. To explore this, we identified disordered regions of MLO-hProt using a pLDDT score less than 70 and partitioned these regions into two categories: drivers (dMLO-hIDR), which actively drive phase separation, and clients (cMLO-hIDR), which are present in MLOs under certain conditions but do not promote phase separation themselves [cite]. Additionally, IDRs from human proteins not associated with MLOs, termed nMLO-hIDR, were included as a control. To enhance statistical robustness, we extended our dataset by incorporating driver proteins from additional species [cite], resulting in the expanded dMLO-IDR dataset. Beyond the pLDDT-based classification, the majority of residues in these datasets are also predicted to be disordered by various computational tools and supported by experimental evidence (Figures S5 and S6).”

> *Recommendation 1: The authors use the terms "evolutionary fitness of IDRs" (abstract and p. 5, for example), "fitness of amino acids" (p. 4), and "quantify the fitness of particular residues at specific sites" (p. 5). It is not clear what is meant by fitness in this context.*

We thank the reviewer for pointing out the ambiguity in the term fitness. To enhance clarity, we have replaced “fitness" with “mutational tolerance" to more directly emphasize the evolutionary conservation of specific residues.

> *Recommendation 2: The authors write (p. 6) "Previous studies have demonstrated a strong correlation between ESM2 scores and changes in free energy related to protein structure stability". While that may be true, it might be worth noting that ESM2 scores report on the effects of mutations and function more broadly than stability because these models have previously been shown to capture conservation effects beyond stability.*

We fully agree with the reviewer’s comment and have revised the main text accordingly. Specifically, the referenced sentence has been revised and relocated, as shown below.

“Our analysis demonstrated that HP1_α_’s structured domains consistently yield low ESM2 scores, reflecting strong mutational constraints characteristic of folded regions. These constraints are further evident in the local LLR predictions, as shown in Figure 2B, where we illustrate the folded region G120-T130. Given the functional importance of preserving the 3D of structured domains, mutations with greater detrimental effects are likely to disrupt protein folding substantially. This interpretation is consistent with previous studies reporting a significant correlation between ESM2 LLRs and changes in free energy associated with protein structural stability [cite].”

> *Recommendation 3: p. 10: The authors write "To exclude sequences that no longer qualify as homologs, we filtered for sequences with at least 20% identity to the reference". How did they decide on 20% and why? And over which residues are these 20% calculated.*

We apologize for the earlier lack of clarity. Sequence alignment was performed using the full-length protein sequences, encompassing both folded and disordered regions. For each sequence, we calculated the percent identity by counting the number of positions, denoted as n, at which the amino acid matches the reference. The percent identity was then computed as n/N, where N represents the total length of the aligned reference sequence. This total includes residues in folded and disordered regions, as well as gap positions introduced during alignment.

We updated the Methods section of the main text to clarify.

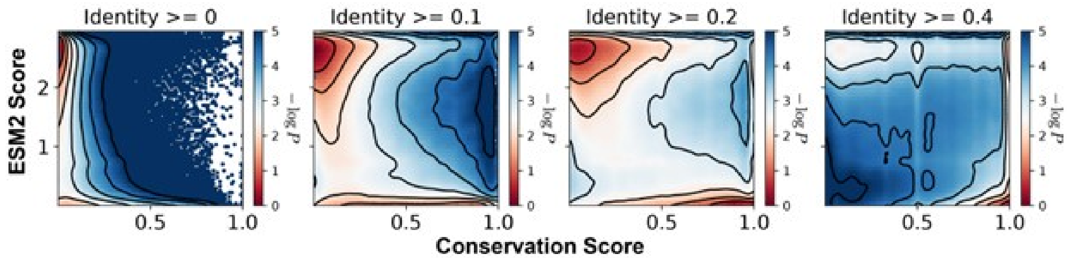
“We performed multi-sequence alignment (MSA) analysis using HHblits from the HH-suite3 software suite [citations], a widely used open-source toolkit known for its sensitivity in detecting sequence similarities and identifying protein folds. HHblits builds MSAs through iterative database searches, sequentially incorporating matched sequences into the query MSA with each iteration. Sequence alignment was performed using the full-length protein sequences, encompassing both folded and disordered regions.

To refine alignment quality by focusing on closely related homologs, we filtered out sequences with ≤ 20% identity to the query, excluding weakly related sequences where only short segments show similarity to the reference. For each sequence, we calculated the percent identity by counting the number of positions, denoted as n, at which the amino acid matches the reference. The percent identity was then computed as n/N, where N represents the total length of the aligned reference sequence. This total includes residues in folded and disordered regions, as well as gap positions introduced during alignment.”

We selected a 20% sequence identity threshold to balance inclusion of true homologs with exclusion of distant matches that may not share functional relevance. To determine this cutoff, we compared identity thresholds of 0%, 10%, 20%, and 40% and examined the resulting distributions of conservation and ESM2 scores across aligned residues for MLO-hProt dataset (Author response image 3). Thresholds of 10%, 20%, and 40% produced qualitatively similar results, with a consistent correspondence between low ESM2 scores and high conservation. Lower thresholds introduced highly divergent sequences that added noise to the alignment, resulting in reduced overall conservation scores. In contrast, higher thresholds excluded homologs with potentially meaningful conservation, particularly in disordered regions where conservation scores tend to be relatively low.

Author response image 3.

Histograms of the ESM2 score and the conservation score, presented in a format consistent with Figure 3B of the main text. The conservation scores were computed using aligned sequences with identity thresholds of ≥0, ≥10%, ≥20%, and ≥40% (left to right). Contour lines represent different levels of −log_P_(CS,ESM2), where P is the joint probability density of conservation score (CS) and ESM2 score. Contours are spaced at 0.5-unit intervals, highlighting regions of distinct density.



> *Recommendation 4: In their description of "motif" searching algorithm (p. 20) I think that the search algorithm would give a different result whether the search is performed N->C or C->N (because the first residue (i) needs to have a score <0.5 but the last (j) could have a score >0.5 as long as the average is below 0.5. Is that correct? And if so, why did they choose an asymmetric algorithm? .*

We thank the reviewer for highlighting the asymmetry in our motif-search algorithm.

To investigate this issue, we repeated the algorithm starting from the C-terminus and compared the resulting motifs with those obtained from the N-terminal scan. We found that the two sets of motifs overlap entirely: each motif identified from the C-terminal direction has a corresponding counterpart from the N-terminal scan. However, the motifs are not identical. The directionality of the search introduces additional amino acids—referred to here as peripheral residues—at the motif boundaries, which differ between the two sets.

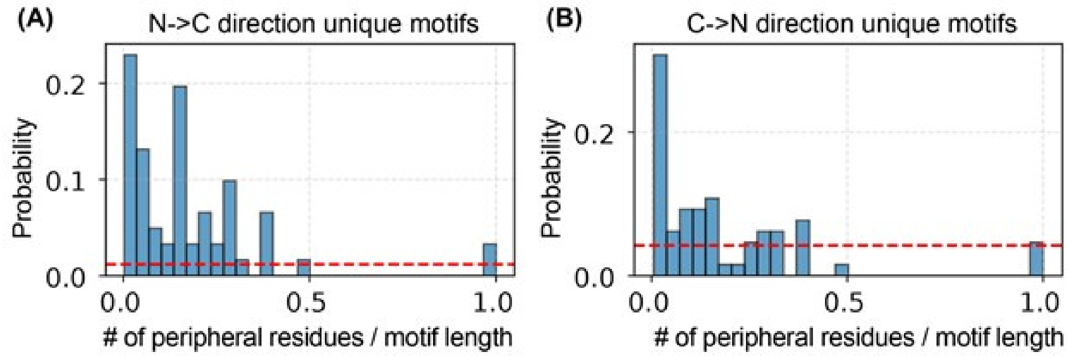
As shown in Author response image 4, the number of peripheral residues is small relative to the total motif length.

To eliminate asymmetry and ambiguity, we have revised our method to perform bidirectional scans—from both the N- and C-termini—and define each motif as the overlapping region identified by both directions. This approach emphasizes the conserved core and avoids the inclusion of spurious terminal residues. The updated procedure is described in Methods: Motif Identification.

"To identify motifs within a given IDR, we implemented the following iterative procedure. Starting from either the N– or C–terminus of the sequence, we first locate the initial residue *i* whose ESM2 score falls within 0.5. From *i*, residues are sequentially appended…"

Author response image 4.

Number of peripheral residues and their relative length to the full-motif length identified from both sides. (A). The unique motifs identified from N-to-C terminal direction. (B) The unique motifs identified from C-to-N terminal direction.



"…in the direction toward the opposite terminus until the segment's average ESM2 score exceeds 0.5; the first residue to breach this threshold is denoted *j*. The segment (*i*,*i*+1,..., *j*−1) is then recorded as a candidate motif. This process repeats starting from *j* until the end of the IDR is reached.

We perform this full procedure independently from both termini and designate the final motif as the intersection of the two candidate-motif sets. This bidirectional overlap strategy excludes terminal residues that might transiently satisfy the average-score criterion only due to adjacent low-scoring regions, thereby isolating the conserved core of each motif. All other residues—those not included in either directional pass—are classified as non-motif regions, minimizing peripheral artifacts."

Accordingly, we have updated the *Supplementary material:* ESM2_motif_with_exp_ref.csv for the new identified motifs commonly exited from both N-terminal and C-terminal searches. Minor changes were observed in the set of motifs as being discussed, but these do not affect the main conclusions. Figures 5C, 5D, and S6 have been revised accordingly.

> ***Reviewer #2:***
>
> *Summary:*
>
> *Unfortunately, I do not believe that the results can be trusted. ESM2 has not been validated for IDRs through experiments. The authors themselves point out its little use in that context. In this study, they do not provide any further rationale for why this situation might have changed. Furthermore, they mention that experimental perturbations of the predicted motifs in in vivo studies may further elucidate their functional importance, but none of that is done here. That some of the motifs have been previously validated does not give any credibility to the use of ESM2 here, given that such systems were probably seen during the training of the model.*

We thank the reviewer for their detailed and thoughtful critique of our manuscript. We recognize the importance of careful model validation, especially in the context of IDRs, and appreciate the opportunity to clarify the scope and rationale of our study. Below, we respond point-by-point to the main concerns.

> *(1) The use of ESM2 is not validated for IDRs, and the authors provide no rationale for its applicability in this context.*

We thank the reviewer for raising this important point.

First, we emphasize that ESM2 is a probabilistic language model trained entirely on amino acid sequences, without any structural supervision. The model does not receive any input about protein structure — folded or disordered — during training. Instead, it learns to estimate the likelihood of each amino acid at a given position, conditioned on the surrounding sequence context. This makes ESM2 agnostic to whether a sequence is folded or disordered; the model's capacity to identify patterns of residue usage arises solely from the statistics of natural sequences.

As such, ESM2 is not inherently biased toward folded proteins, even though previous studies have demonstrated its usefulness in identifying conserved and functionally constrained residues in structured domains [3–9]. These findings support the broader utility of language models for uncovering evolutionary constraints — and by extension, suggest that similar signatures could exist in IDRs, particularly if they are under functional selection.

Indeed, if certain residues or motifs in IDRs are conserved due to their importance in biological processes (e.g., phase separation), we would expect such selection to be reflected in sequence-based features, which ESM2 is designed to detect. The model's applicability to IDRs, then, is a natural extension of its core probabilistic architecture.

To further evaluate this, we carried out an independent in silico validation using multiple sequence alignments (MSAs). This analysis allowed us to compute the evolutionary conservation of individual amino acids without any reliance on ESM2. We then compared these conservation scores to ESM2 scores and found a strong correlation between the two. This provides direct, quantitative support for the idea that ESM2 is capturing biologically meaningful sequence constraints — even in disordered regions.

While we agree that experimental testing would ultimately provide the most compelling validation, we believe that our MSA-based comparison constitutes a strong and arguably ideal computational validation of the model's predictions. It offers an orthogonal measure of evolutionary pressure that confirms the biological plausibility of ESM2 scores.

We added the following text in the introduction to highlight the applicability of ESM2 to IDRs.

"While protein language models have been widely applied to structured proteins, it is important to emphasize that these models themselves are not inherently biased toward folded domains. For example, the Evolutionary Scale Model (ESM2) [cite] is trained as a probabilistic language model on raw protein sequences, without incorporating any structural or functional annotations. It operates by estimating the likelihood of observing a given amino acid at a particular position, conditioned on the entire surrounding sequence context. This unsupervised learning paradigm enables ESM2 to capture statistical patterns of residue usage and evolutionary constraints without relying on explicit structural information. Thus, the success of ESM2 in modeling fitness landscapes of folded proteins reflects the model's ability to learn sequence-level constraints imposed by natural selection — a property that is equally applicable to IDRs if those regions are also under functional selection. Indeed, protein language models are increasingly been used to analyze variant effects in IDRs [cite]."

> *(2) There is no experimental validation of the ESM2-based predictions in this study.*

We agree that experimental validation would provide definitive support for the utility of ESM2 in IDRs, and we explicitly state this as a limitation in the revised manuscript as quoted below.

"Limitations: Despite the promising findings, our study has several limitations. Most notably, our analysis is purely computational, relying on ESM2-derived predictions and sequence-based conservation without accompanying experimental validation. While the strong correlation between ESM2 scores and evolutionary conservation provides compelling evidence that the identified motifs are functionally constrained, the precise biological roles of these motifs remain uncharacterized. ESM2 is well-suited for highlighting regions under selective pressure, but it does not provide mechanistic insights into how conserved motifs contribute to specific molecular functions such as phase separation, molecular recognition, or dynamic regulation. Determining these roles will require targeted experimental investigations, including mutagenesis and biophysical characterization.”

In addition, we revised the manuscript title from “Protein Language Model Identifies Disordered, Conserved Motifs Driving Phase Separation" to “Protein Language Model Identifies Disordered, Conserved Motifs Implicated in Phase Separation". This revision softens the original claim to better reflect the absence of direct experimental evidence for the motifs' role in phase separation.

However, we also emphasize that the goal of our study is not to claim definitive predictive power, but rather to explore whether ESM2-derived mutational profiles align with known biological features of IDRs — and in doing so, to generate new, testable hypotheses.

In addition, while no in vivo experiments were performed, our study does include an in silico validation step, as detailed in the response to the previous comment. The strong correlation between ESM2 scores and conservation scores provides direct support for the utility of ESM2 in identifying residues under evolutionary constraint in disordered regions.

> *(3) The overlap between predicted motifs and known ones may be due totraining data leakage.*

We respectfully clarify that training data leakage is not possible in this case, as ESM2 is trained using unsupervised learning on raw protein sequences alone. The model has no access to experimental annotations, functional labels, or knowledge of which motifs are involved in phase separation. It only models statistical sequence patterns derived from evolutionarily observed proteins.

Therefore, any agreement between ESM2-derived predictions and previously validated motifs arises not from memorization of experimental data, but from the model's ability to learn meaningful sequence constraints from the natural distribution of proteins.

> *(4) The authors should revamp the study with a testable predictive framework.*

We respectfully suggest that a full revamp is not necessary or appropriate in this context.

As outlined in our previous responses, we believe that certain misunderstandings about the nature and capabilities of ESM2 may have influenced the reviewer's assessment.

Importantly, both Reviewer 1 and Reviewer 3 express strong support for the significance and novelty of this work, and recommend publication following minor revisions.

In this context, we believe the manuscript provides a useful contribution as a first step toward understanding disordered regions using language models, and that it has value even in the absence of direct experimental testing. We have now better positioned the manuscript in this light, clarified limitations, and suggested concrete next steps for follow-up research.

We hope these clarifications and revisions address the reviewer's concerns, and we thank them again for helping us strengthen the framing, rigor, and clarity of our study.

> ***Reviewer #3:***
>
> *Summary:*
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> *This is a very nice and interesting paper to read about motif conservation in protein sequences and mainly in IDRs regions using the ESM2 language model. The topic of the*

> *paper is timely, with strong biological significance. The paper can be of great interest to the scientific community in the field of protein phase transitions and future applications using the ESM models. The ability of ESM2 to identify conserved motifs is crucial for disease prediction, as these regions may serve as potential drug targets. Therefore, I find these findings highly significant, and the authors strongly support them throughout the paper. The work motivates the scientific community towards further motif exploration related to diseases.*
>
> *Strengths:*
>
> *(1) Revealing conserved regions in IDRs by the ESM-2 language model.*
>
> *(2) Identification of functionally significant residues within protein sequences, especially in IDRs.*
>
> *(3) Findings supported by useful analyses.*

We appreciate the reviewer's thoughtful words and support for our work.

> *Weaknesses:*
>
> *(1) Lack of examples demonstrating the potential biological functions of these conserved regions.*

As detailed in the Response to Recommendation 6, we conducted additional analyses to connect the identified conserved regions with their biological functions.

> *(2) Very limited discussion of potential future work and of limitations.*

We have substantially revised the Conclusions and Discussion section to provide a detailed analysis of the study's limitations and to propose several directions for future research, as elaborated in our Response to Recommendation 5 below.

> *Recommendation 1: The authors describe the ESM2 score such that lower scores are associated with conserved residues, stating that "lower scores indicate higher mutational constraint and reduced flexibility, implying that these residues are more likely essential for protein function, as they exhibit fewer permissible mutational states." However, when examining intrinsically disordered regions (IDRs), which are known to drive phase separation, I observe that the ESM2 score is relatively high (Figure 3C, pLDDT < 50, and Supplementary Figure S2). Could the authors clarify how this relatively high score aligns with the conservation of motifs that drive phase separation?*

We thank the reviewer for this insightful comment. We would like to clarify that most amino acids in the IDRs are not conserved, even for IDRs that contribute to phase separation. Only a small set of amino acids in these IDRs, which we term as motifs, are evolutionarily conserved with low ESM2 scores. Therefore, the ESM2 scores exhibit bimodal distribution at high and low values, as shown in Figures 4A and 4C of the manuscript. When averaged over all the amino acids, the mean ESM2 scores, plotted in Figure 3C, are relatively high due to dominant population of non-conserved amino acids.

> *Recommendation 2: The authors mention: "We first analyzed the relationship between ESM2 and pLDDT scores for human Heterochromatin Protein 1 (HP1, residues 1-191)". I appreciate this example as a demonstration of amino acid conservation in IDRs. However, it is questionable whether the authors could provide some more examples to support amino acid conservation particularly within the IDRs along with lower ESM2 score (e.g, Could the authors provide some additional examples of "conserved*

> *disordered" regions in various proteins which are associated with relatively low ESM2 score as appear in Figure 2A).*

We thank the reviewer for this valuable suggestion. We want to kindly noted that the conserved residues on IDRs are prevalent as indicated in Figures 2D and 3B. To further illustrate the prevalence of “conserved disordered” regions, we generated ESM2 versus pLDDT score plots for the full dMLO–hProt dataset (82 proteins) in Figure S2. In these plots, residues with pLDDT ≤ 70 are highlighted in blue to denote structural disorder (dMLO-hIDR), and these disordered residues with ESM2 score ≤ 1.5 are shown in purple to indicate conserved disordered segments.

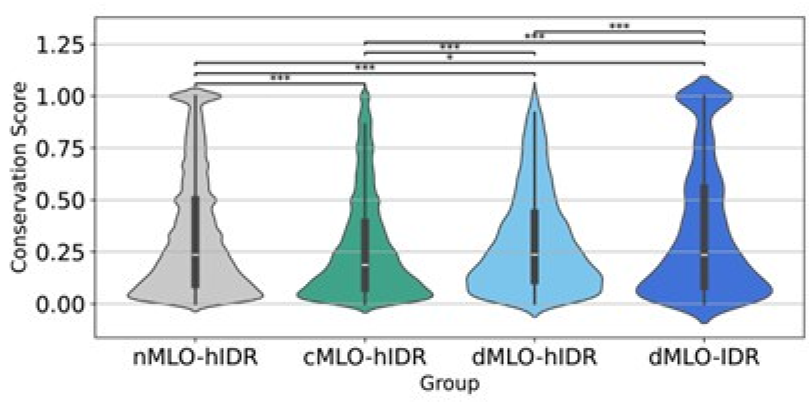
> *Recommendation 3: Could the authors plot a Violin conservation score plot for Figure 4A to emphasise the relationship between ESM2 scores and conservation scores of disordered residues?*

We thank the reviewer for this suggestion. We included a violin plot illustrating the distribution of conservation scores for disordered residues across all four IDR groups, shown in Author response image 5. Consistent with the findings in Figure 4A, the phase separation drivers (dMLO-hIDR and dMLOIDR) exhibit a higher proportion of conserved amino acids compared to the client group (cMLOhIDR).

We also note that the nMLO-hIDR group may contain conserved residues due to functions unrelated to MLO formation, which could contribute to the higher observed levels of conservation in this group.

Author response image 5.

Violin plots illustrating the distribution of conservation scores for disordered residues across the nMLO–hIDR, cMLO–hIDR, dMLO–hIDR, and dMLO–IDR datasets. Pairwise statistical comparisons were conducted using two-sided Mann–Whitney U tests on the conservation score distributions (null hypothesis: the two groups have equal medians). P-values indicate the probability of observing the observed rank differences under the null hypothesis. Statistical significance is denoted as follows: ***: p < 0.001; **: p < 0.01; *:p < 0.05;



> *Recommendation 4: It will be appreciated if the authors could add to Figure 4 Violin plots, a statistical comparison between the different groups.*

We thank the reviewer for this valuable suggestion. We included the p-values for Figures 4A and 4C to quantify the statistical significance of differences in the distributions.

Most comparisons are highly significant ($p < 0.001$), while the largest p-value ($p = 0.089$) between the conservation score of driving and non-participating groups (Figure 4C) still suggests a marginally significant trend.

> *Recommendation 5: Could the authors expand more on potential future research directions using ESM2, given its usefulness in identifying conserved motifs? Specifically, how do the authors envision conserved motifs will contribute to future discoveries/applications/models using ESM (e.g, discuss the importance of conserved motifs, especially in IDRs motifs, in protein phase transition prediction in relation to diseases).*

We thank the reviewer for this insightful comment. To further assess the functional relevance of the conserved motifs, we incorporated pathogenic variant data from ClinVar [10, 11] to evaluate mutational impacts. As shown in Figure S12A and B, a substantial number of pathogenic variants in MLO-hProt proteins are associated with low ESM2 LLR values. This pattern holds for both folded and disordered residues.

Moreover, we observed that variants located within motifs are more frequently pathogenic compared to those outside motifs (Figure S12C). In the main text, motifs were defined only for driver proteins; however, the available variant data for this subset are limited (6 data points). To improve statistical power, we extended motif identification to include both client and driver human proteins, following the same methodology described in the main text. Consistent with previous findings, variants within motifs in this expanded set are also more likely to be pathogenic. These results further support the functional importance of both low ESM2-scoring residues and the conserved motifs in which they reside.

The following text was added in the *Discussion* section of the manuscript to discuss these results and outline future research directions.

"Several promising directions could extend this work, both to refine our mechanistic understanding and to explore clinical relevance. One avenue is testing the hypothesis that conserved motifs in scaffold proteins act as functional stickers, mediating strong intermolecular interactions. This could be evaluated computationally via free energy calculations or experimentally via interaction assays. Deletion of such motifs in client proteins may also reduce their partitioning into condensates, illuminating their roles in molecular recruitment.

To explore potential clinical implications, we analyzed pathogenicity data from Clin-Var [10, 11]. As shown in Figure S12A, single-point mutations with low LLR values—indicative of constrained residues—are enriched among clinically reported pathogenic variants, while benign variants typically exhibit higher LLR values. Moreover, mutations within conserved motifs are significantly more likely to be pathogenic than those in non-motif regions (Figure S12B). These findings highlight the potential of ESM2 as a first-pass screening tool for identifying clinically relevant residues and suggest that the conserved motifs described here may serve as priorities for future studies, both mechanistic and therapeutic."

Moreover, the functional significance of conserved motifs, particularly their implications in disease and pathology, warrants further investigation. As an initial analysis, we incorporated ClinVar pathogenic variant data [citation] to assess mutational effects within our datasets. As illustrated in Figure R12A, single-point mutations with low LLR values are enriched among clinically reported pathogenic variants, whereas benign variants are more commonly associated with higher LLR values. Notably, mutations within conserved motifs are substantially more likely to be pathogenic compared to those in non-motif regions. These findings highlight the potential of ESM2 as a firstpass tool for identifying residues of clinical relevance. The conserved motifs identified here may be prioritized in future studies aimed at elucidating their biological roles and evaluating their viability as therapeutic targets.

> *Recommendation 6: The authors mention: "Our findings provide strong evidence for evolutionary pressures acting on specific IDRs to preserve their roles in scaffolding phase separation mechanisms, emphasizing the functional importance of entire motifs rather than individual residues in MLO formation." They also present a word cloud of functional motifs in Figure 5D. Although it makes sense that evolutionarily conserved motifs, especially within the IDRs regions, act as functional units, I think there is no direct evidence for such functionality (e.g., examples of biological pathways associated with IDRs and phase separation). Hence, there is no justification to write in the figure caption: "ESM2 Identifies Functional Motifs in driving IDRs" unless the authors provide some examples of such functionality. This will even make the paper stronger by establishing a clear connection to biological pathways, and hence these motifs can serve as potential drug targets.*

We thank the reviewer for this insightful suggestion. We have replaced “functional motifs" with “conserved motifs" in the figure caption.

Identifying the precise biological pathways associated with the conserved motifs is a complex task and a comprehensive investigation lies beyond the scope of this study. Nonetheless, as an initial effort, we explored the potential functions of these motifs using annotations available in DisProt (https://disprot.org/).

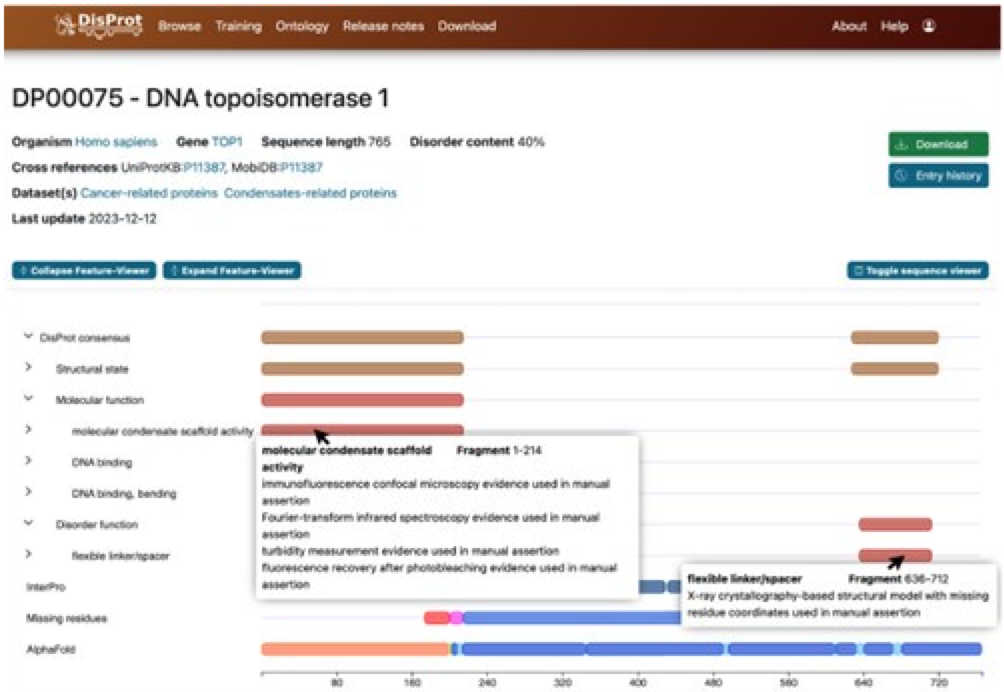
DisProt is the leading manually curated database dedicated to IDPs, providing both structural and functional annotations. Expert curators compile experimentally validated data, including definitions of disordered regions, associated functional terms, and supporting literature references. Author response image 6 presents a representative DisProt entry for DNA topoisomerase 1 (UniProt ID: P11387), illustrating its structural and biological annotation.

For each motif, we located the corresponding DisProt entry and assigned a functional annotation based on the annotated IDR from which the motif originates. We emphasize that this functional assignment should be regarded as an approximation. Because experimental annotations often pertain to the entire IDR, regions outside the motif may also contribute to the reported function.

Nevertheless, the annotations provide valuable insights.

Author response image 6.

Screenshot of information provided by the DisProt database. Detailed annotations of biological functions and structural features, along with experimental references, are accessible via mouse click.



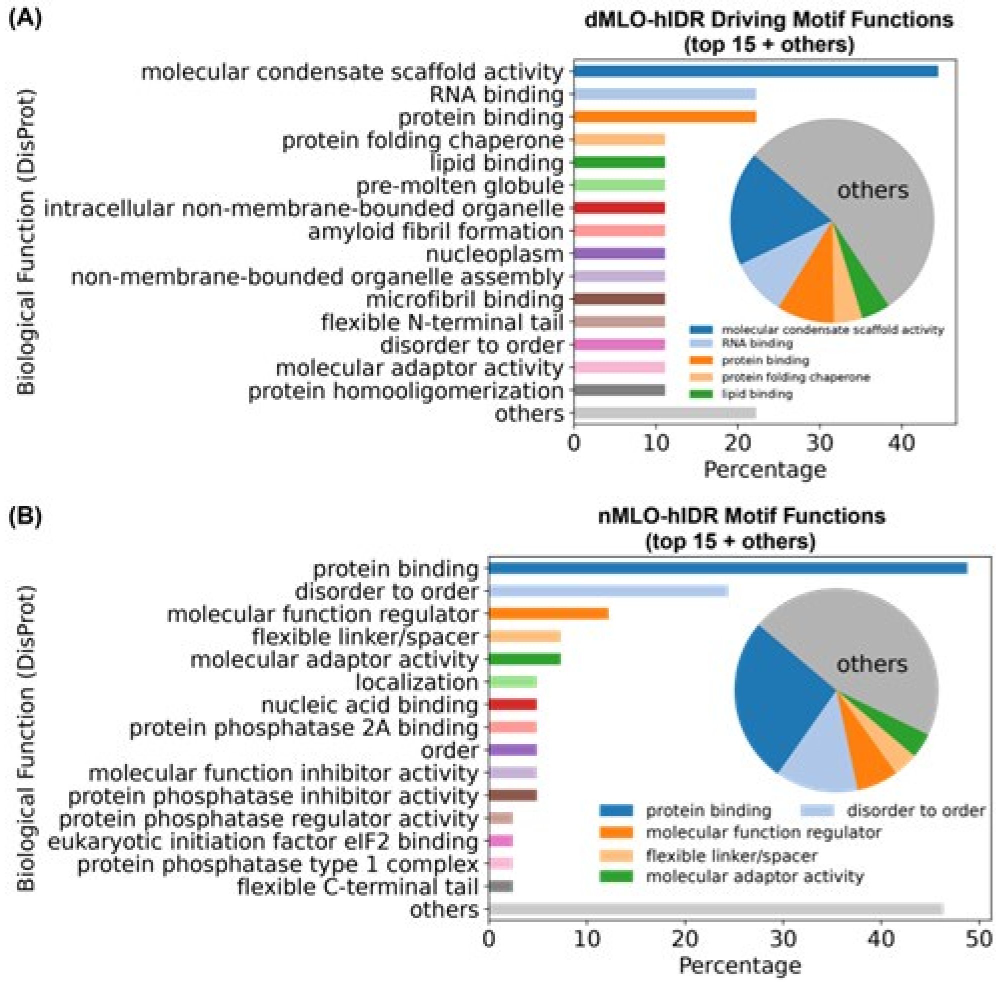
Approximately 50% of ESM2-predicted IDR motifs lack functional annotations. Among those that are annotated, motifs from the dMLO-IDR dataset are predominantly associated with “molecular condensate scaffold activity,” followed by various biomolecular binding functions (Author response image 7A). These findings support the role of these motifs in MLO formation.

For comparison, we applied the same identification procedure (described in Methods: Motif Identification) to motifs from the nMLO-hIDR dataset. In contrast to the dMLO-IDR motifs, these exhibit a broader range of annotated functions related to diverse cellular processes. Collectively, these results suggest that motifs identified by ESM2 are aligned with biologically relevant functions captured in current databases.

Finally, as illustrated in Figure S12 and discussed in the Response to Recommendation 5, variants occurring within identified motifs are more likely to be pathogenic than those in non-motif regions, further underscoring their functional importance.

Author response image 7.

Biological functions of ESM2-predicted motifs. (A) Distribution of biological functions associated with all identified motifs from dMLO-IDR driving groups. (B) Distribution of biological functions associated with all identified motifs from nMLO-hIDR groups.



> *Recommendation 7: In Figure 2C the authors present FE (I assume this is free energy), some discussion about the difference in the free energy referring to the "a" region is missing (i.e. both "Folded" and "Disordered" regions are associated with low ESM score but with low and high free energy (FE), respectively.*

We thank the reviewer for the comments. FE indeed abbreviates free energy. To improve clarify and avoid confusion, we have updated all figure captions by replacing "FE" with "−log*P*" to explicitly denote the logarithm of probability in the contour density plots.

We used "a" in Figures 2C and 2D to refer to regions with low ESM2 scores, which appears a local minimum in both plots. Since most residues in folded regions are conserved, region a has lower free energy than region b in Figure 2C. On the other hand, as most residues in disordered regions are not conserved, as we elaborated in *Response to Recommendation 1*, region a has lower population and higher free energy than region b.

To avoid confusion, we have replaced "a" and "b" in Figure 2D with "I" and "II".

> *Recommendation 8: Figure S2: It would be useful to plot the same figure for structured and disordered regions as well.*

We are not certain we fully understood this comment, as we believe the requested analysis has already been addressed. In Figure S2, we used the AlphaFold2 pLDDT score to represent the structural continuum of different protein regions, where residues with pLDDT > 70 (red and lightred bars) are classified as structured, while those with pLDDT ≤ 70 (blue and light-blue bars) are classified as disordered.

> *Minor suggestion 1: Could the authors clarify the meaning of the abbreviation "FE" in the colorbar of the contour line? I assume this is free energy.*

We have updated all contour density plot figure captions by replacing “FE” with “−log*P*” to explicitly denote the logarithm of probability.

> *Minor suggestion 2: In Figure 2A - do the authors mean "Conserved folded" instead of just "Folded"? If so, could the authors indicate this?*

We thank the reviewer for this comment. The ESM2 scores indeed suggest that, within folded regions, there may be multiple distinct groups exhibiting varying degrees of evolutionary conservation. However, as our primary focus is on IDRs, we chose not to investigate these distinctions further.

Figure 2A illustrates a randomly selected folded region based on AlphaFold2 pLDDT scores.

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